K.R. MANGALAM UNIVERSITY

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ATTENDANCE SHORTAGE CALCULATOR

First Year Project Synopsis
Submitted by

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Project Overview

Attendance is a critical academic requirement in educational institutions, with a minimum percentage mandatory for exam eligibility. However, students often fail to meet this requirement due to delayed awareness of attendance shortages.

The Attendance Shortage Calculator is a web-based application designed to help students track their attendance in real time, calculate attendance percentage, predict shortages in advance, and determine how many future classes are required to meet the minimum attendance criteria. This system helps reduce academic risk and last-minute stress.

Specific Objectives

- To calculate the current attendance percentage accurately.
- To determine whether a student meets the minimum attendance requirement.
- To predict attendance shortage in advance.
- To calculate the number of classes required to recover attendance deficit.
- To help students plan attendance and avoid academic penalties.

Key Features

- Subject-wise attendance calculation.
- Automatic attendance percentage computation.
- Attendance eligibility check (e.g., 75% rule).
- Attendance shortage prediction.
- Recovery class calculation
- Simple and user-friendly dashboard

Project Use cases & Scope

- Students tracking attendance regularly.
- Early detection of attendance shortages.
- Academic planning before examinations.
- Scope.
- Applicable for schools, colleges, and universities.
- Can be extended to integrate with ERP systems.
- Suitable for web and mobile platforms in the future

Project Flow

User

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Web Interface (HTML, CSS, JavaScript)

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Backend Logic (Python – Flask)

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Database (MySQL / MongoDB)

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Attendance Analysis & Result Display

Data & Resources

- Data Used.
- User-entered attendance data:
- Total lectures conducted.
- Lectures attended.
- Resources.
- Web browser for application access.
- Backend server using Flask.
- Databases for data storage (MySQL).
- No sensitive or external institutional data is used.

Methodology

- User inputs subject-wise attendance details.
- System calculates attendance percentage.
- Percentage is compared with minimum required threshold.
- If attendance is insufficient:
    - System predicts shortage
    - Calculates number of classes needed to recover
    - Results are displayed through a dashboard
- Tools & Technologies Used
    - HTML
    - CSS
    - JavaScript
    - Python (Flask Framework)
    - MySQL

Expected Results & Impact

- Accurate attendance calculation.
- Early warning for attendance shortage.
- Clear recovery plan for students.
- Impact.
- Reduced academic stress.
- Improved attendance awareness.
- Better academic planning.
- Fewer last-minute penalties and detentions

Project Timeline

- Phase 1: Requirement Analysis – 1 Week
- Phase 2: UI Design – 1 Week
- Phase 3: Backend Development – 2 Weeks
- Phase 4: Database Integration – 1 Week
- Phase 5: Testing & Refinement – 1 Week

THANK YOU